What Makes Small Decision Models Transfer? Schema Augmentation and Data-Mixture Trade-offs in a 350M Decision Model

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***Abstract*—Generative language models are increasingly misused as structured decision routers, paying $O(T)$ sequential decoding cost and returning uncalibrated confidences. System-One decision models instead map a state and a typed schema to calibrated probabilities in a single forward pass. We adapt LiquidAI LFM2.5-350M, a hybrid convolution-attention model, into a decision engine with a cross-attention JointSchemaHead trained via QLoRA. Across seven controlled training runs and six evaluation kernels we find: (1) leave-one- taxonomy-out transfer is substantial but partial—44% on BANKING77 and 55.6% on CLINC150 against 2%/17% chance, versus 81.3%/72.9% in-distribution means over three seeds; (2) an earlier 36% $\rightarrow$ 83% augmentation jump was a context-truncation confound, not semantic robustness: with row-matched budgets, canonical training wins in-distribution (86%/84% vs 79%/70%) while augmentation buys transfer and memory savings; (3) calibration does not transfer—ECE rises from 0.026 in-distribution to 0.366 under shift—yet error-detection (AUROC 0.87 $\rightarrow$ 0.73) survives, and realistic autonomous coverage collapses from 63% to near zero. All checkpoints, dumps, and logs are archived for reproduction.**

Keywords*—System-One decision models, model calibration, transfer learning, small language models, hybrid architectures, selective classification

I. Introduction

Enterprise automation—fraud routing, ticket triage, policy enforcement—needs decisions that are fast, syntactically valid, and uncertainty-aware. The dominant practice prompts an autoregressive model to emit JSON, incurring three costs: (i) $O(T)$ sequential decoding (1.5–4 s per decision with KV-cache traffic), (ii) non-zero schema-violation rates under constrained budgets, and (iii) systematic overconfidence from preference-based alignment, which makes raw softmax scores unusable for risk-controlled automation.

System-One decision models [1]–[3], inspired by dual-process theory [4], separate deciding from generating: a state S and a typed schema Q are mapped to a joint distribution over options in one forward pass ($O(1)$ decoding steps), with probabilities trained to be calibrated. Open questions remain for sub-billion models: does schema augmentation drive transfer to unseen taxonomies, what does data-mixture composition trade off at fixed capacity, and does calibration survive distribution shift?

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We study these on LiquidAI LFM2.5-350M, a hybrid model with convolutional sequence mixers and grouped-query attention, fitted with a cross-attention JointSchemaHead via QLoRA. Our contributions, each backed by controlled runs with fixed test sets, are:

1) **Measured transfer:** leave-one-taxonomy-out evaluation gives 44% on BANKING77 and 55.6% on CLINC150 (chance: 2%/17%) versus in-distribution means of 81.3%/72.9% over three seeds.
2) **A confound found and fixed:** an earlier 36% $\rightarrow$ 83% augmentation jump was context truncation (2048 tokens cutting wide schemas), not semantic robustness. Row-matched canonical training wins in-distribution (86%/84% vs 79%/70%); augmentation buys transfer plus memory savings at a small in-domain tax.
3) **Calibration under shift:** ECE rises from 0.026 in-distribution to 0.366 on the held-out taxonomy while error-detection AUROC only falls from 0.87 to 0.73; autonomous coverage at 5% risk collapses from 63% to near zero. Temperature scaling fit in-distribution does not transfer.
4) **Artifacts:** seven training runs and six evaluation kernels with checkpoints, per-item dumps, and logs archived for reproduction.

Section 2 positions the work, Section 3 describes data, Section 4 the architecture and losses, Section 5 the protocol, Section 6 results, and Section 7 concludes.

II. Related Works

A. System-One Decision Models

TypeSafe's Jev frames automation as decision routing with honest probabilities [1]. Cloudflare's Clef (27B) and Clef-flash (9B) pair frozen Qwen backbones with a joint schema head trained with label-smoothed cross-entropy, Brier penalty, and an RLCD secondary objective; weights are open under Apache-2.0 [2]. Convai's Laya shows sub-500M bidirectional encoders reaching single-digit millisecond latency [3]. LiquidAI's open d1 family (d1-omni-600M, same LFM lineage as our backbone) provides the closest size-class reference [5]. We tributeevaluate against vendor-reported numbers only, clearly labeled, since rerunning 9–27B models is outside a T4 protocol.

B. Which RLCD?

We use RLCD as *Reinforcement Learning for Calibrated Decisions* (TypeSafe/Cloudflare: RL with a proper-scoring-rule reward). This is unrelated to Yang et al.'s *Reinforcement Learning from Contrastive Distillation* [6], which generates preference pairs without a calibration objective. Shared acronym only. Analysis of RLCD variance and the *calibrate, then reinforce* paradigm appears in [7], [8].

C. Calibration and Selective Classification

A scoring rule is strictly proper if truth-telling uniquely maximizes the expected score [9]. The Brier score [10] and its reliability–resolution–uncertainty decomposition [11] are our main tools. Guo et al. showed modern networks are overconfident and temperature scaling helps [12]. Selective classification formalizes abstention via coverage and selective risk [13]; our coverage-at-risk metric follows that line.

D. Zero-Shot Decision Benchmarks

TypedBench proposes accuracy, paraphrase robustness, calibration against a finite-sample noise floor, and selective prediction for decision models [14]; we borrow its paraphrase-grid and selective protocol. System Switch shows accuracy, ECE, and error-detection AUROC are distinct quantities, with deferral gain tracking AUROC ($\rho = 0.87$) [15]—exactly the dissociation we observe under shift. Small-model precedents include distillation from System-One labels [16] and matched-token-budget 360M finetunes [17]. Head-to-head benchmarking finds small decision models beating small generative ones on knowledge tasks while losing on math [18], which bounds where our 350M claims apply: intent and routing taxonomies, not reasoning benchmarks. Companion efforts cover moderation confidence benchmarking [19], meta-encoding and knowledge-tracing variants [20], [21]. Option-order sensitivity motivates schema augmentation as a treatment [22].

III. Dataset and Preprocessing

A. Mixture Composition

Training uses a 13-source mixture (20 400 sampled rows) plus 9 600 consensus-soft rows: Open-Jev release-v2 (3 600) and mailroom-control-v1 (2 400), LeWiDi conv_abuse and csc sarcasm (1 200 each, annotator-share soft labels), and GoEmotions (1 200 emotion-fraction rows). Intent taxonomies (BANKING77, CLINC150) are upweighted $3\times$; twelve public intent and reasoning sources fill the remainder. Custom rows are mapped explicitly: Open-Jev `options` lists (`id: description`) become criteria with annotator-share probabilities; LeWiDi soft labels map onto `true`/`false` (abusive/non-abusive) and six sarcasm levels; GoEmotions multi-rater rows become per-emotion fractions. All rows are rendered into canonical decision records.

B. Schema Augmentation

Augmentation (treatment E) applies option subsampling and shuffling (wide taxonomies capped at 24 options), instruction paraphrasing, and derived binary (noul) questions. Treatment A disables all augmentation (canonical schemas). With augmentation the same row budget yields more decisions (39 560 vs 31 754), a residual we disclose rather than hide.

TABLE I. Training mixture (20 400 sampled + 9 600 consensus-soft rows).

Group	Rows	Labels
12 public sources	20 400	hard
Intent share	$3\times$ weight	hard
Open-Jev (2 subsets)	3 600 + 2 400	soft
LeWiDi (2 configs)	1 200 + 1 200	annotator shares
GoEmotions fractions	1 200	soft

C. Decontamination and Splits

All rows are decontaminated against canonical test splits with 13-gram shingle matching before training. Items are split train/calibration/test with fixed seeds; temperature scaling is fit on calibration only, verdicts on held-out tests. For leave-one-taxonomy-out (LOTO) runs, every row, label definition, and augmentation variant of the held-out taxonomy is removed from training. Render skips are counted, not silent: typical runs skip $\sim$800 of $\sim$43 000 decisions, dominated by over-context schemas and non-binary noul criteria, and the counts are logged per run for audit.

IV. Proposed Methodology

A. Hybrid Backbone

We adapt `LiquidAI/LFM2.5-350M` [23]: 16 layers (10 double-gated short-range convolutional LIV blocks plus 6 grouped-query attention blocks), 32K context, 65 536 vocabulary. Convolutions give constant-memory state caching while attention preserves associative recall—an unexplored substrate for decision heads, which prior work mounts on dense bidirectional encoders [3] or decoder-only transformers [2]. The choice is deliberate, not incidental: intent routing leans on short-range lexical matching (double-gated convolutions cache it in constant memory) plus label-set associative recall (a few GQA layers suffice), so a hybrid trunk fits edge serving where a dense transformer of equal quality would not. Our feasibility result (Section 6) is therefore also a deployment result: the same weights that route intents at 81% fit a T4 with 4-bit quantization and no KV-cache growth from generation.

B. Canonical Input Encoding

Each instance (state S, questions $Q = \{q_j\}$ with types $\tau_j \in \{\text{choice}, \text{score}, \text{noul}\}$ and option criteria O_j) is rendered deterministically: state text followed by per-question blocks (ID, type, instruction, named options with descriptions). When length exceeds budget, truncation cuts strictly the tail of S—schema text is never cut, and rows whose schema alone overflows are skipped with a logged reason. This invariant is load-bearing for Section VI.

C. JointSchemaHead

For final hidden states $\mathbf{H} \in \mathbb{R}^{B\times L\times D}$, projected to width $W = 256$, candidate options attend to full-sequence memory (2 cross-attention layers), fields are conditioned through 4 decoder layers with a learned type embedding $\mathbf{e}_\tau$, and logits combine a lexical prior with a joint MLP over $[\mathbf{f}_q; \mathbf{o}; \mathbf{f}_q \odot$

$\mathbf{o}; |\mathbf{f}_q - \mathbf{o}|]$ gated by $\sigma(g)$. Requesting a plain LM from the Unsloth API attaches this Clef-design head by default, confirming the lineage. Inference is one forward pass with per-question softmax.

D. Parameter-Efficient Adaptation

Full finetuning of 350M parameters per recipe is wasteful when the backbone already models language. Low-Rank Adaptation (LoRA) [24] freezes W_0 and learns $\Delta W = BA$ with $B \in \mathbb{R}^{d\times r}$, $A \in \mathbb{R}^{r\times k}$, $r \ll \min(d,k)$, cutting trainable parameters by orders of magnitude while keeping the forward pass $h = W_0 x + \frac{\alpha}{r} BAx$. QLoRA [25] additionally quantizes the frozen backbone to 4-bit, so a 350M model trains in T4 memory with adapters and optimizer states as the only full-precision footprint. We use rank $r = 16$, $\alpha = 16$, dropout 0, targets on all projection layers (`q_proj`, `k_proj`, `v_proj`, `out_proj`, `in_proj`, `w1--w3`), Unsloth gradient checkpointing, and seeded initialization. The backbone stays frozen; only adapters and the decision head train. The trainer splits the optimizer automatically: backbone adapters at the configured LR (2×10^{-4}) and head parameters at 1×10^{-4}.

E. Training Pipeline (Verified Against Unsloth 2026.10.3)

Mixture. Rows are sampled by weight (intent taxonomies 3×) with 2× headroom so decontamination never forces resampling; sources flagged eval-only are rejected from training by the library. *Augmentation* (treatment E) draws per-row variations with default probabilities: option rename 0.3 (letter/code ids, meaning moved into descriptions), instruction paraphrase 0.4, instruction drop 0.1, derived binary questions 0.5, question-id rename 0.3, state-as-text 0.3, field shuffle on, and wide taxonomies subsampled to max_options $= 24$ around the gold label. Treatment A uses canonical rows. *Items.* Each decision becomes $\{\text{input_ids}, \text{markers}, \text{qtype}, \text{target}, \text{label}\}$; unrenderable rows are skipped with counted reasons (e.g. non-binary noul criteria, over-context schemas), and over-length inputs truncate state tails only. *Split.* Holdout takes whole rows (never splitting a row's questions across train/holdout) at 10% capped at 400 decisions; we halve it into calibration and test with a seeded shuffle. *Optimize.* Per-epoch evaluation and checkpointing keep best eval-loss (epoch-3 overfitting was consistent, so epoch 2 always wins) with patience-2 early stopping. *Calibrate.* Dual temperatures are fit by L-BFGS with two-fold cross-fitting: reported numbers score each half of the rows with temperatures fit on the other half, so calibration error is never measured with in-sample temperatures. *Evaluate/predict* use the served temperatures; `predict` additionally returns per-option probabilities for proper-scoring analysis.

F. Multi-Objective Loss

$$\mathcal{L} = \mathcal{L}_{\text{LS-CE}} + \lambda_{\text{Brier}}\,\mathcal{L}_{\text{Brier}} + \lambda_{\text{ord}}\,\mathcal{L}_{\text{Ordinal}}, \tag{1}$$

with label smoothing [26] ($\tilde{y} = (1-\epsilon)y + \epsilon/K$, $\epsilon = 0.05$), a Brier penalty against one-hot targets punishing overconfident errors, and an ordinal partial credit for `score` types. (The trainer's reference-KL term defaults to zero and is unused here.)

V. Experimental Setup

A. Protocol

Seven training runs (3 seeds of the base recipe, 2 LOTO, 2 ablations) on NVIDIA T4 via Kaggle kernels: AdamW, backbone LR 2×10^{-4}, head LR 1×10^{-4}, cosine decay, effective batch 32, 3 epochs keeping best eval-loss (epoch-3 overfitting was consistent, so the best checkpoint is always epoch 2). Evaluation uses identical test sets everywhere (BANKING77/CLINC150 500 items, typed-decisions 400): canonical accuracy plus a 3-variant paraphrase grid (wording and option-order perturbations), bootstrap 95% intervals (100 resamples on GPU runs), measured per-item latency on T4, and per-item probability dumps (1 400 rows each) scored with Brier, NLL, 10-bin ECE, error-detection AUROC, and coverage at 5% risk.

Algorithm 1 Three-layer evaluation with paraphrase and probability audit.

Build canonical benches once per tokenizer (typed 400, intent 500 each)
Before: backbone + random head (chance reference)
After: trained checkpoint; time per-item latency
Robustness: 2 wording/order variants per bench, report median+range
Uncertainty: bootstrap 95% CI over items (GPU runs)
Calibration: cross-fitted ECE on cal split; `predict` dump per-item probabilities → Brier/NLL/AUROC/coverage offline

Vendor numbers (Unsloth Qwen, Cloudflare Clef) are cited as labeled context, never as rerun baselines; the size-matched reference is d1-omni-600M from the same LFM lineage [5]. Every run archives adapters, head, merged weights, and `run_meta.json` in its kernel output; lightweight results (`eval_results.json`, probability dumps, metric tables) are versioned, so each number below traces to a logged artifact.

TABLE II. Hyperparameters (all seven training runs).

Setting	Value
Backbone	LFM2.5-350M, 4-bit, frozen
LoRA rank / alpha / dropout	16 / 16 / 0
Backbone / head LR	2×10^{-4} / 1×10^{-4}
Schedule / warmup	cosine / 5%
Effective batch	32 (8×4; 4×8 full schemas)
Epochs / selection	3, best eval-loss, patience 2
Context	4 096 tokens
Seeds	3407 / 41 / 42

VI. Results and Discussion

A. In-Distribution: Three Seeds

On identical tests, the base recipe gives typed-decisions 55.9/53.3/55.6 (mean 54.9), BANKING77 76.8/83.4/83.8 (mean 81.3), and CLINC150 68.2/76.6/73.8 (mean 72.9). Seed spread on BANKING77 (76.8 vs ∼83.5) is exactly why single-seed reporting is insufficient. Untrained-head baselines sit at chance (31%/2%/17%).

TABLE III. IN-DISTRIBUTION ACCURACY ON IDENTICAL TESTS (500/500/400 ITEMS).

Seed	typed	BANKING77	CLINC150
3407	55.9	76.8	68.2
41	53.3	83.4	76.6
42	55.6	83.8	73.8
Mean	54.9	81.3	72.9

B. Transfer: Leave-One-Taxonomy-Out

With BANKING77 fully held out, accuracy reaches **44% [38–47%]** against 2% chance (paraphrase median 42.8%, range 42.2–43.6); with CLINC150 held out, **55.6%** (paraphrase 55.6–57.0) against 17% chance. Transfer is substantial but partial versus 81.3%/72.9% in-distribution.

Transfer is wording-robust, not prompt luck: a 3-variant grid (instruction rewording plus option-order shuffling) moves LOTO BANKING77 only across 42.2–43.6% (median 42.8%) and LOTO CLINC150 across 55.6–57.0%, while in-distribution A holds 86.0–86.4% on BANKING77. Ranges this tight mean the effect lives in schema semantics, not phrasing.

TABLE IV. PARAPHRASE GRID: MEDIAN [RANGE] OVER 3 WORDING/OPTION-ORDER VARIANTS.

Model	BANK77	CLINC150
LOTO-banking	42.8 [42.2–43.6]	—
LOTO-clinc	—	56.0 [55.6–57.0]
A canonical	86.0 [86.0–86.4]	84.6 [84.4–84.8]
E augmented	79.4 [79.0–80.2]	70.0 [69.8–70.4]

TABLE V. LOTO TRANSFER VS IN-DISTRIBUTION MEANS (ACCURACY %).

Setting	BANKING77	CLINC150
Held out (transfer)	44 [38–47]	55.6
In-distribution mean	81.3	72.9
Chance	2	17

C. Ablation: The Truncation Confound

Row-matched canonical (A) vs full-augmentation (E) training gives:

TABLE VI. ABLATION A VS E (28 000 ROWS EACH; DECISIONS 31 754 VS 39 560).

Treatment	typed	BANKING77	CLINC150
A canonical	54% [51–56]	**86%** [81–88]	**84%** [81–88]
E augmented	59% [56–61]	79% [75–82]	70% [65–73]

An earlier 36% → 83% jump attributed to augmentation was a context truncation confound: training at 2 048 tokens cut wide schemas, and augmentation's option subsampling merely made them fit. With fitting context, canonical training wins in-distribution; augmentation buys transfer plus memory savings (full 77/150-option schemas OOM at batch 8 on T4) at a small in-domain tax. Residual confounds disclosed: +25% decisions for E (biasing toward E) and batch 4 vs 8 at equal effective size 32.

Overfitting dynamics are strikingly consistent: eval-loss falls through epoch 2 then spikes at epoch 3 in every fully-logged run, so best-checkpoint selection always restores epoch 2 (Table VII).

TABLE VII. EVAL-LOSS AT EPOCHS 2 AND 3 (BEST CHECKPOINT UNDERLINED).

Run	Epoch 2	Epoch 3
seed41	0.72	0.90
seed42	0.62	1.01
LOTO-banking	0.59	0.72
LOTO-clinc	0.64	0.87
Ablation A	0.76	0.95
Ablation E	0.57	0.65

D. Calibration Under Shift

On BANKING77, ECE rises from **0.026** in-distribution to **0.366** held-out (Brier 0.281 → 0.876); autonomous coverage at 5% risk collapses from 63% to 0.6%. Temperature scaling fit in-distribution does not transfer. Yet error-detection AUROC falls only from 0.87 to 0.73: confidence still separates right from wrong answers, confirming that accuracy, calibration, and sensitivity are distinct quantities [15]. Measured T4 latency is 56–249 ms/item depending on schema width.

TABLE VIII. CALIBRATION PER BENCH: LOTO (SHIFT) VS E (IN-DISTRIBUTION).

Model	Bench	Acc	ECE	AUROC	Cov@5%
LOTO	typed	59	0.053	0.64	3
LOTO	banking*	43	0.366	0.73	0.6
LOTO	clinc	65	0.176	0.82	29
E	typed	59	0.121	0.62	0.1
E	banking	80	0.026	0.87	63
E	clinc	70	0.090	0.85	40

E. External Context

Table IX places our accuracy beside vendor-reported numbers. Metric and protocol boundaries differ per row (Decision Index reports macro-F1 over full tests; Unsloth reports accuracy under its own mixture; ours is accuracy on identical 500/500/400 tests), so no cross-row ranking is claimed. Two observations survive the boundary: (i) a deployed 421M system (Laya) sits near chance on intent taxonomies while our 350M reaches 81–86%, i.e. recipe dominates size in this class; (ii) 9–27B systems lead absolutely (90–97%), as expected, with Cloudflare's own rows flagged self-reported by the board.

TABLE IX. CONTEXT TABLE. OURS: ACCURACY ON IDENTICAL TESTS. STARRED ROWS: VENDOR-REPORTED UNDER OWN PROTOCOLS AND METRICS (ACC VS MACRO-F1).

Model	Params	BANK77	CLINC150	typed
Ours mean (acc)	350M	81.3	72.9	54.9
Ours canonical A (acc)	350M	86	84	54
Qwen3.5-0.8B* (acc)	0.8B	74	76	73
Clef* (F1)	27B	94.2	97.4	—
Clef-flash* (F1)	9B	90.9	66.8	—
Jev* (F1)	closed	79.7	89.3	—
Kev 9B* (F1)	9B	84.8	79.0	—
Laya* (F1)	421M	14.3	3.2	—

F. Capacity Frontier

Typed-decisions plateaus at 54–59% across all seven runs while Qwen3.5-0.8B reports 73%: unlike single wide taxonomies (handled well even at 150 classes), its records demand up to five interdependent judgments per state (risk, review need, routing) that stretch 16 layers at width 1 024. The gap is architectural capacity, not recipe, and it bounds deployment scope to single-question routing rather than joint multi-field reasoning.

G. Discussion

Fixed 350M capacity forces mixture trade-offs: doubling the typed-decisions share shifted BANKING77 +2 but CLINC150 −7 without lifting either, i.e. weights relocate capacity rather than create it. For practitioners: fit schemas in context first (truncation dwarfs recipe effects), buy augmentation for transfer and memory rather than in-domain points, and never trust in-distribution temperatures under shift. Deploy with an escalation policy

$$\hat{y}(x) = \arg\max_k p_k(x) \text{ if } \max_k p_k(x) \geq \tau \text{ else ESCALATE}, \tag{2}$$

with τ set from AUROC-validated coverage analysis (our models sustain 63% autonomy in-distribution at 5% risk but near zero under shift, so τ must be re-fit per deployment distribution). Limits: A/E are single-seed with a batch-size confound (4×8 vs 8×4 at equal effective size 32, forced by full-schema OOM); rename/paraphrase/derived types are not yet separated, so the transfer credit cannot be attributed to one mechanism; no same-protocol Qwen rerun exists (vendor numbers are labeled context only); OOS/OOD slices and macro-F1 leaderboard slices are future work; and the `lfm1.0` license text is still unread, which must precede any release claim. None of these touch the three headline findings, each of which rests on fixed test sets with disclosed residuals.

VII. CONCLUSION

A 350M hybrid convolution-attention backbone, fitted with a joint schema head, is a viable System-One decision engine: 81.3%/72.9% in-distribution means over three seeds, 44%/55.6% held-out-taxonomy transfer, and honest, reproduced numbers throughout. Our main methodological finding is a cautionary one: a dramatic augmentation effect was context truncation in disguise, and calibration fit in-distribution collapses under shift even as error detection survives. Artifacts—seven training runs, six evaluation kernels, per-item dumps, and logs—are archived for reproduction. Checkpoints (adapters, head, merged weights) plus `run_meta.json` live in per-run kernel outputs; lightweight results (`eval_results.json`, probability dumps, metric tables) are versioned in `docs/eval/`, and every paper number traces to a logged run. The immediate next steps are separating augmentation types (rename vs paraphrase vs derived) to attribute the transfer credit, adding OOS/OOD and macro-F1 leaderboard slices, and replacing single post-hoc temperatures with shift-aware calibration (per-domain temperatures or a proper-loss RLCD stage [7]). Future work separates augmentation types, extends OOS/OOD and leaderboard slices, and explores shift-aware calibration.

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